

COMPANION
to

ChatGPT Teaches TikZ

Chapter 27: Advanced Nodes and Anchors (A)

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27.1 What This Companion Adds

Chapter 27 treated a node as a geometric object with named reference points. This Companion develops an anchor-first design method for stable alignment, corner attachments, typographic baselines, predictable routing, and layouts that survive changes in label length.

Design

An anchor is a contract between an object and its surroundings. Choose the reference point that expresses the relationship before considering a manual shift.

27.2 An Anchor Vocabulary

Anchor or option	Purpose
anchor=west	Places a new node by its left boundary.
(A.east)	Refers to the right boundary of an existing node.
north, south	Refers to horizontal side centers.
north east, south west	Refers to corner anchors.
base	Aligns the text baseline.
mid	Aligns the vertical middle of lowercase text.
inner sep	Sets padding between contents and boundary.
outer sep	Adds clearance outside the node boundary.
minimum width, minimum height	Establishes a minimum box size.

27.3 Placement Anchors and Reference Anchors

These two uses of anchors answer different questions.

```
\node[draw,anchor=west] (A) at (0,0)
  {A long label};
\draw (A.east)--++(1cm,0);
```

The option `anchor=west` says which part of the new node is placed at $(0,0)$. The coordinate `(A.east)` later refers to a point on the finished node.

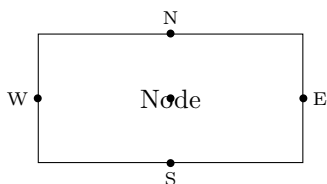
Common Mistake

Do not read `anchor=east` as “put the node east of the coordinate.” It places the node’s east anchor at the coordinate, so most of the node extends to the west.

27.4 Seeing the Principal Anchors

Temporary marks are useful while learning or debugging node geometry.

See



Remove diagnostic marks after the anchor choice has been verified.

27.5 Aligning Unequal Nodes

Center alignment makes unequal widths extend by unequal amounts on both sides. Align the boundary that readers should perceive as common.

```
\node[draw,anchor=west] at (0,2) {Short};  
\node[draw,anchor=west] at (0,1) {A longer label};  
\node[draw,anchor=west] at (0,0)  
  {A substantially longer label};
```

For a column of right-aligned boxes, use `anchor=east`. For a row with a common upper edge, use `anchor=north`.

27.6 Two Anchors Define an Attachment

A corner annotation uses one anchor from the existing object and one from the new object.

```
\node[draw,minimum size=14mm] (A) {Main};  
\node[draw,circle,inner sep=1.5pt,  
      anchor=south west]  
  at (A.north east) {1};
```

The main node supplies `A.north east`; the small node places its own `south west` there. This is a complete geometric description with no shift to tune.

27.7 Box Alignment and Text Alignment

Use `center`, `north`, and related anchors when the shapes must align. Use `base` when the text must share a typographic baseline.

```
\node[anchor=base] at (0,0) {$x$};  
\node[anchor=base] at (1.3,0) {$x^2$};  
\node[anchor=base] at (2.8,0) {$x_i$};  
\node[anchor=base] at (4.3,0) {$\sum_i x_i$};
```

The `mid` anchor can look more balanced for rows of ordinary lowercase text. It is based on the middle of the letter x rather than the baseline.

Design

Align boxes when the boundary carries the structure. Align baselines when the reader should perceive the contents as one line of text or mathematics.

27.8 Boundary Routing

Automatic connections between node names usually meet the shapes correctly. Explicit anchors are preferable when the direction of entry or exit is part of the diagram.

```
\draw[->] (A.east)--(B.west);  
\draw[->] (C.south)--(D.north);
```

For an orthogonal route, combine an anchor with the path operators.

```
\draw[->] (A.east)---+(8mm,0) |- (B.east);
```

The initial horizontal segment gives the route clearance before it turns.

27.9 Padding Changes Geometry

`inner sep` enlarges the distance from the contents to the boundary, so it also moves boundary anchors. `minimum width` and `minimum height` impose lower bounds but allow longer contents to enlarge the node.

```
\tikzset{stage/.style={
  draw,rounded corners,
  minimum width=30mm,
  minimum height=9mm,
  inner sep=4pt
}}
```

Do not compensate for a poor size rule by shifting every edge. Correct the node geometry at the style level.

27.10 Anchors with Relative Positioning

The positioning library determines the overall relationship; anchors refine the boundary connection.

```
\node[stage] (A) {Source};
\node[stage,right=18mm of A] (B) {Destination};
\draw[->] (A.east)--(B.west);
```

This division of labor is easier to revise than a collection of absolute coordinates and shifts.

27.11 Worked Project: A Stable Proof Pipeline

27.11.1 Stage 1: Define the Common Geometry

```
\tikzset{
  stage/.style={draw,rounded corners,
    minimum width=32mm,minimum height=9mm,
    align=center,inner sep=4pt},
  transition/.style={-{Stealth},thick}
}
```

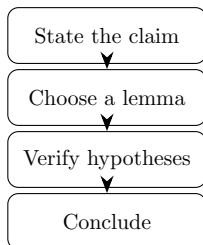
27.11.2 Stage 2: Align the Left Edges

```
\node[stage,anchor=west] (A) at (0,3) {State the claim};  
\node[stage,anchor=west] (B) at (0,2) {Choose a lemma};  
\node[stage,anchor=west] (C) at (0,1) {Verify hypotheses};  
\node[stage,anchor=west] (D) at (0,0) {Conclude};
```

27.11.3 The Complete Picture

```
\begin{tikzpicture}
  \node[stage,anchor=west] (A) at (0,3)
    {State the claim};
  \node[stage,anchor=west] (B) at (0,2)
    {Choose a lemma};
  \node[stage,anchor=west] (C) at (0,1)
    {Verify hypotheses};
  \node[stage,anchor=west] (D) at (0,0)
    {Conclude};
  \draw[transition] (A.south)--(B.north);
  \draw[transition] (B.south)--(C.north);
  \draw[transition] (C.south)--(D.north);
\end{tikzpicture}
```

See



Lengthen any stage label. The west boundary remains aligned and the vertical connections continue to meet the north and south anchors.

27.12 Debugging Workshop

27.12.1 The Node Extends in the Wrong Direction

The placement anchor is probably opposite the intended fixed boundary.

27.12.2 Boxes Align but Formulas Do Not

Use a baseline-related anchor when typographic alignment is the real goal.

27.12.3 An Edge Attaches to an Unexpected Side

Specify the relevant boundary anchors explicitly.

27.12.4 A Longer Label Breaks the Layout

Inspect the minimum-size rule and relative spacing before adding a local shift.

27.13 Exercises

1. Align three unequal nodes by their left boundaries.
2. Align three unequal nodes by their upper boundaries.
3. Attach a small label to the lower-right corner of a node.
4. Align four mathematical labels by their baselines.
5. Connect the east side of one node to the west side of another.

6. Route an arrow around the right side of two vertical nodes.
7. Change one label and test whether every alignment survives.
8. Replace two manual shifts in an earlier diagram with anchors.

27.14 Solutions

27.14.1 Exercises 1–4

```
\node[draw,anchor=west] at (0,2) {One};
\node[draw,anchor=west] at (0,1) {Longer two};
\node[draw,anchor=north] at (3,2) {Three};
\node[anchor=north east] at (A.south east) {note};
\node[anchor=base] at (0,0) {$x_i$};
```

27.14.2 Exercises 5–7

```
\draw[->] (A.east)--(B.west);
\draw[->] (C.east)--++(8mm,0) |- (D.east);
```

Exercise 7 lengthens a label without adding a repair shift. Exercise 8 chooses anchors that state the alignments already intended by the old source.

27.15 ChatGPT Workshop

Ask ChatGPT to identify the geometric reference points that a layout needs.

1. Distinguish placement anchors from references to existing nodes.
2. Replace small alignment shifts with edge or corner anchors.
3. Choose between box alignment, baseline alignment, and mid alignment.
4. Diagnose an arrow that changes sides after a label becomes longer.
5. Propose a label-length test for every important node style.

27.16 Final Checklist

Before continuing to Chapter 28, check that you can

1. distinguish placing an anchor from referencing one;
2. align unequal nodes by a meaningful boundary;
3. define corner attachments with two anchors;
4. choose typographic or geometric alignment;
5. route edges through explicit boundary anchors;
6. account for node padding and minimum size; and
7. test a layout with longer labels.